EXPERIMENT NO:03

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier, plot_tree

# Training dataset
data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rain', 'Rain',
                'Rain', 'Overcast', 'Sunny', 'Sunny', 'Rain',
                'Sunny', 'Overcast', 'Overcast', 'Rain'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool',
                    'Cool', 'Cool', 'Mild', 'Cool', 'Mild',
                    'Mild', 'Mild', 'Hot', 'Mild'],
    'Humidity': ['High', 'High', 'High', 'High', 'Normal',
                 'Normal', 'Normal', 'High', 'Normal', 'Normal',
                 'Normal', 'High', 'Normal', 'High'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Weak',
             'Strong', 'Strong', 'Weak', 'Weak', 'Weak',
             'Strong', 'Strong', 'Weak', 'Strong'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes',
             'No', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'No']
}

df = pd.DataFrame(data)
```

```
print("Training Dataset:")
print(df)

# Convert categorical data into numeric form
X = pd.get_dummies(df.drop('Play', axis=1))
y = df['Play']

# Build ID3-style decision tree using entropy
model = DecisionTreeClassifier(criterion='entropy', random_state=0)
model.fit(X, y)

# Predict a new sample
sample = pd.DataFrame([{
    'Outlook': 'Sunny',
    'Temperature': 'Cool',
    'Humidity': 'High',
    'Wind': 'Strong'
}])

sample_encoded = pd.get_dummies(sample).reindex(
    columns=X.columns, fill_value=0
)

prediction = model.predict(sample_encoded)

print("\nNew Sample:")
```

```
print(sample)
print("Predicted Play:", prediction[0])

# Graph 1: Decision tree
plt.figure(figsize=(16, 8))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=model.classes_,
    filled=True,
    rounded=True
)
plt.title("ID3 Decision Tree")
plt.show()

# Graph 2: Class distribution
plt.figure(figsize=(6, 4))
df['Play'].value_counts().plot(
    kind='bar',
    color=['skyblue', 'orange']
)
plt.title("Play Tennis Class Distribution")
plt.xlabel("Play")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.grid(axis='y')
```

plt.show()

OUTPUT:

